EvoDiff: generating proteins in sequence space

Protein generation with evolutionary diffusion:
sequence is all you need

Alamdari et al. — bioRxiv v2 (2024); eLife Reviewed Preprint v1 (2026)

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Which version are we discussing?

Requested paper: bioRxiv 10.1101/2023.09.11.556673v2, posted 4 November 2024. **Accessible full source:** eLife 15:RP112029, Reviewed Preprint v1, 24 August 2026; DOI: 10.7554/eLife.112029.1. The eLife record identifies the 2024 preprint and includes full methods, supplementary figures/tables, and public reviews. It is not a final version-of-record publication. These notes use that accessible scientific record. Direct bioRxiv access returned 403, so a line-by-line v2 comparison was unavailable.

eLife version history and status; figures carry 556673v2 identifiers.

Question: can sequence be the design space?

Structure-first design

Desired geometry → generated backbone → compatible sequence. Powerful when a fixed 3D configuration defines the task; structural training data cover a selected subset of proteins.

Sequence-first design

Natural sequences → learned sequence distribution → constrained sequence generation. Can include intrinsically disordered regions (IDRs), whose function may depend on dynamic ensembles.

Central contribution: one model supports arbitrary fixed residues and missing regions, without training a new model for each design task.

Introduction; Fig. 1. This is a capability claim, not a claim that structure is irrelevant.

Two kinds of discrete corruption

Both operate on amino-acid tokens, rather than continuous coordinates.

	Masking: OADM	Substitution: D3PM
Toy forward path	`M A K L` $\rightarrow$ `M ? K L` $\rightarrow$ `? ? K L` $\rightarrow$ `? ? ? ?`	`M A K L` $\rightarrow$ `M V K I` $\rightarrow$ `W V E I` $\rightarrow$ random residues
Terminal state	All masks; $T = L$	Uniform amino acids; $T = 500$
Reverse process	Fill one masked position; keep filled residues	Repeatedly resample residue identities

? denotes a mask. Paths are illustrations, not actual model samples.
OADM is also discrete diffusion; "masked versus discrete" is imprecise.

Fig. 1B–C; Methods: D3PM and OADM.

OADM: generate in any order

Length $L = 4$; choose a random decoding order, e.g. $(3, 1, 4, 2)$.

$$\texttt{????} \rightarrow \texttt{??K?} \rightarrow \texttt{M?K?} \rightarrow \texttt{M?KL} \rightarrow \texttt{MAKL}$$

Each step predicts the next residue from *all currently visible residues*. For a permutation σ,

$$p_\theta(x \mid \sigma) = \prod_{j=1}^{L} p_\theta(x_{\sigma_j} \mid x_{\sigma_1}, \ldots, x_{\sigma_{j-1}}).$$

Training samples a masking level and random visible subset, then predicts masked residues with a reweighted cross-entropy objective. A conventional MLM trains at a restricted corruption regime. OADM trains across the full range, including nearly all-mask contexts used at generation.

Methods: OADM; relation to LRAR and MLMs. Equation shows a fixed-order factorization.

D3PM: replace tokens, then learn to repair them

For a one-hot residue x_0, use a categorical transition matrix Q_t:

$$q(x_t \mid x_0) = \mathrm{Cat}(x_0 \bar{Q}_t), \qquad \bar{Q}_t = Q_1 Q_2 \cdots Q_t.$$

1. Corrupt residues independently conditional on the clean sequence.
2. A context-aware neural network predicts the original token distribution.
3. Combine that prediction with the known corruption process to sample a reverse step.

Uniform transitions ignore residue relationships. **BLOSUM62-based** transitions favor plausible substitutions early, approaching uniform noise later. D3PM can revise a position repeatedly; OADM commits after decoding it.
In this study, the simpler OADM task generally works better.

Methods: D3PM; $T = 500$, variational objective ($\lambda = 0$); Fig. S25.

Same framework, two inputs and two backbones

	EvoDiff-Seq	EvoDiff-MSA
Input	Single amino-acid sequence	Alignment of related sequences
Architecture	Dilated ByteNet-style CNN	MSA Transformer
Scale explored	38M and 640M parameters	100M parameters
Training data	About 42M UniRef50 training sequences	372,296 OpenFold training MSAs
Training input	Crops up to 1,024 residues	64 rows, up to 512 columns
Selected model	640M OADM	OADM, Max-diversity training

MSA subsampling: random rows or greedy MaxHamming selection.
Alignment supplies family context; it does not require a family-specific retraining run.

Results: discrete diffusion; Methods: datasets, subsampling, architectures; Tables S1–S2.

Conditioning is just leaving the right tokens visible

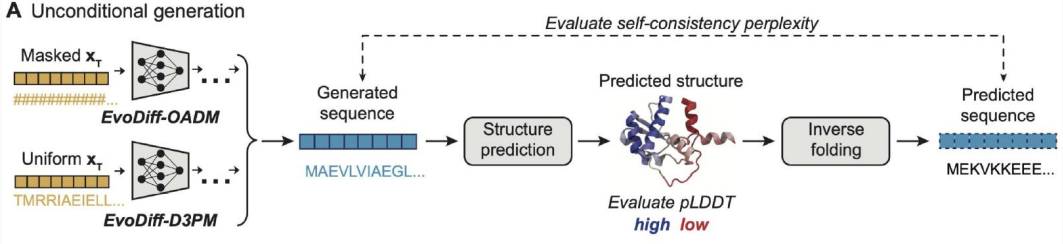
Task	Fixed context → generated content
Inpainting	`M A ? ? ? L V` → replace only the missing region
Motif scaffolding	`? ? C ? ? H ? ?` → generate a scaffold around fixed motif residues
Family generation	Fixed homolog rows + all-mask query → new query sequence

No finetuning is needed for these uses of OADM. Amino-acid identities are fixed; their 3D geometry is not guaranteed.

For MSA scaffolding, homologous context supplies much more information than an isolated motif string.

Fig. 1D; conditional generation and motif-scaffolding Methods. Toy sequences are illustrative.

Evaluation separates plausibility, coverage, and function



pLDDT: confidence of OmegaFold's predicted structure; higher is better for a folded-design screen. **scPerplexity:** inverse-folding likelihood of the generated sequence given its predicted structure; lower suggests consistency. Neither is a direct measurement of folding, expression, binding, or disorder.

The paper therefore also tests distributional properties and selected designs in the lab.

Fig. 2A, cropped; Methods: structural plausibility. ProteinMPNN provides an additional inverse-folding check.

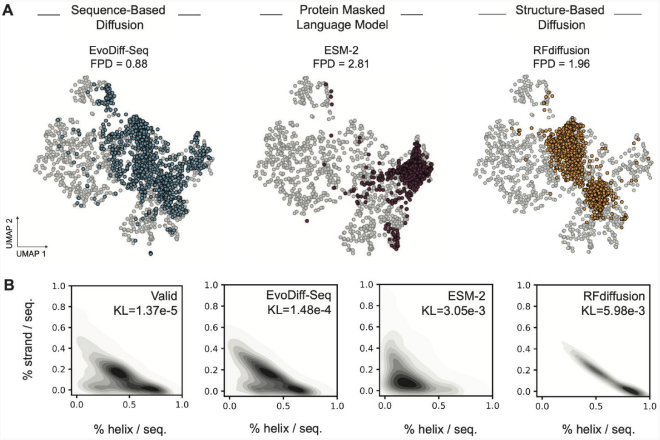
Unconditional quality: OADM is useful, LRAR stays strong

Model	pLDDT ↑	scPerplexity ↓	FPD ↓
Natural test	68.25	8.04	0.10
D3PM-BLOSUM 640M	44.14	11.86	1.53
D3PM-Uniform 640M	41.65	12.29	1.35
OADM 640M (EvoDiff-Seq)	44.46	11.53	0.88
LRAR 640M	48.71	10.99	0.63
ESM-2 650M	50.70	14.02	6.79

Means over 1,000 generated sequences for structural metrics; ESM-IF scPerplexity. Natural-test FPD compares test and validation distributions. Takeaway: OADM improves over the tested D3PMs, but does not dominate unconditional baselines. Its main advantage is flexible conditioning.

Tables S1, S3; units: pLDDT 0–100, perplexity/FPD unitless. No standard deviations shown here.

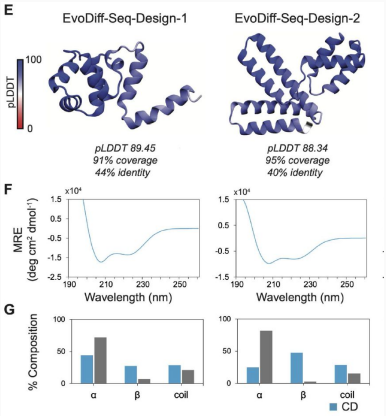
Distribution coverage improves over these MLM/structure baselines



1,000 generated samples per model. RFdiffusion backbones are inverse-folded by ESM-IF. ProtT5 embedding distance and predicted secondary-structure composition suggest broader natural-like coverage for EvoDiff. Interpretation: proxy coverage, not experimentally verified functional diversity. "Other" secondary structure does not establish disorder.

Fig. 3; Methods: FPD and secondary-structure features. LRAR has lower FPD than EvoDiff (Table S1).

Unconditional wet lab: four characterized designs



Selection funnel

1,000 generated sequences

↓ pLDDT > 70 nomination

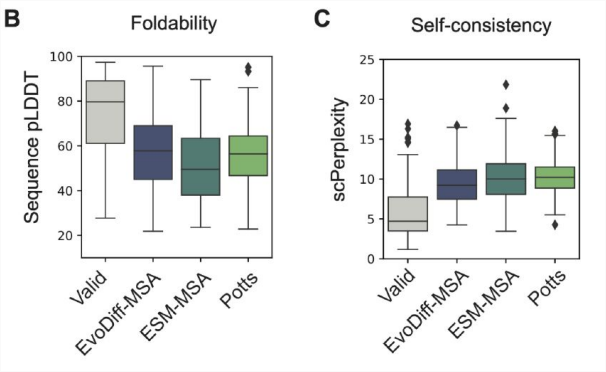
25 designs tested

↓

4 expressed and characterized by circular dichroism (CD). CD supports mixed secondary structure. It does not prove the predicted atomic fold or biochemical activity. 4/25 is a selected-candidate result, not a random-generation success rate.

Fig. 2E–G, cropped: two of four designs; expression/CD Methods; Supplemental Table 1. Many high-pLDDT candidates failed.

MSA conditioning generates plausible family members



Test: mask and regenerate the query from 250 validation MSAs; retain homolog rows. Compare to ESM-MSA sampling and family-trained Potts models. EvoDiff-MSA improves predicted folding/consistency while generating queries distant from the nearest aligned member. TM-score compares *predicted* original and generated structures. This is family-conditioned plausibility, not a direct function assay.

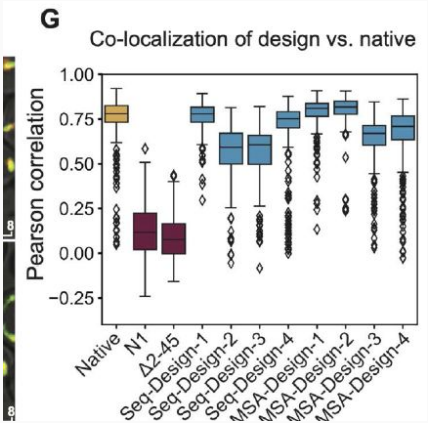
Fig. 4B–C, cropped; Table S4; Methods: evolution-guided generation.

IDR inpainting: does the surrounding protein give context?

Input: a protein with a predicted IDR masked out. EvoDiff-Seq sees the remaining sequence; EvoDiff-MSA also sees homologous context. **Evaluation:** DR-BERT disorder predictions for 100 generated examples per condition; compare natural and random IDRs. Both models preserve predicted disorder patterns better than random replacements. MSA designs retain more similarity to the original IDR. Critical distinction: predicting a disordered region and demonstrating a functional disordered region are separate tests.

Fig. 5A–C; Figs. S11–S13; Methods: generation of IDRs.

Cox15: a selected targeting-signal pipeline works in cells



Replace yeast Cox15 residues 1–45 with designed N-terminal signals. 200 candidates → DR-BERT + MitoFates ranking → 8 tests (4 Seq, 4 MSA). All 8 show localization consistent with native

Cox15, quantified by sfGFP/mScarlet colocalization. Controls: native signal, deletion Δ2–45, and inactive bulk-property-matched N1 signal.
183–313 cells/design; biological replicate in Fig. S17.

Fig. 5G, cropped; Fig. 5D–F; Figs. S14–S17. Cells are not independent design replicates.

What the Cox15 experiment establishes

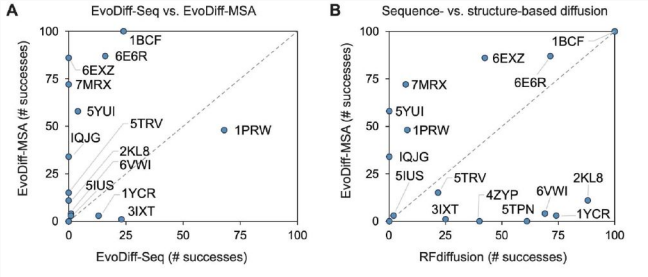
Demonstrated: the full generation, predictor-selection, and testing pipeline can yield functional mitochondrial targeting signals in this protein context. **Not established by this assay:**

1. Experimental disorder of the generated signals.
2. Broad success across unrelated IDR functions or proteins.
3. An 8/8 hit rate for unfiltered EvoDiff samples.

The mature Cox15 sequence strongly identifies a mitochondrial protein. MSA designs also receive orthologous information. Both make this a favorable context for inferring targeting signals.

Fig. 5 and selection Methods; eLife public review #2, point 3. Interpretation follows the evidence scope.

Scaffolding: sequence constraints can recover 3D motifs



17 unique motif problems; 100 generations per problem. Some motifs are discontinuous. Success: predicted motif Cα RMSD < 1Å and global pLDDT > 70. At least one success:

EvoDiff-Seq: **8/17**
EvoDiff-MSA: **13/17**. MSA outperforms RFdiffusion on 6 problems; strengths vary by target.

Fig. 6A–B, cropped; Tables S5–S6. Figure omits problems with no successes; denominator above retains all 17.

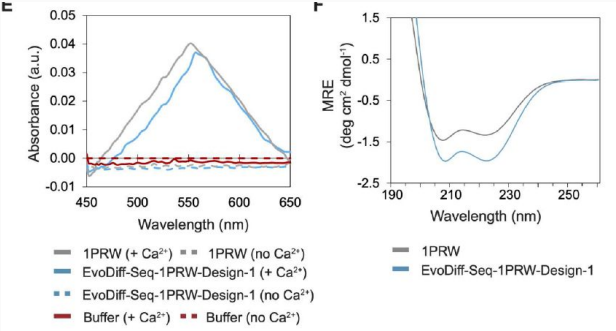
Reading the RFdiffusion comparison fairly

1. **Inputs differ.** EvoDiff-Seq uses residue identities; EvoDiff-MSA also uses homologs. RFdiffusion receives motif coordinates.
2. **Generation pipelines differ.** RFdiffusion needs inverse folding; scaffold lengths differ across methods. Its scaffolding model is finetuned.
3. **Success is computational.** A predicted motif geometry is not a functional binding or catalysis measurement.

ESM-MSA performs similarly to EvoDiff-MSA on scaffolding, a task aligned with its masked training objective (Table S6). Supported conclusion: sequence conditioning is a viable, complementary route. These comparisons do not establish universal method superiority.

Scaffolding Methods; Results; Tables S5–S6; eLife public review #2.

Calmodulin scaffold: expression is easier than binding

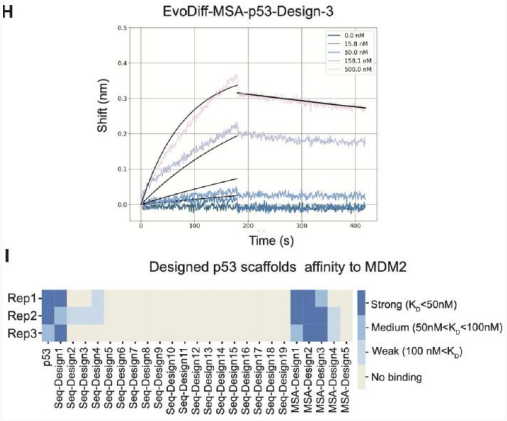


300 generations
100 each: Seq, MSA-Max, MSA-Rand. 13 nominated by predicted metrics plus manual feasibility filtering.
12 expressed successfully.
1 showed a Ca^{2+}-dependent spectral shift. Native calmodulin and buffer $\pm$ Ca^{2+} controls; independent triplicate binding experiments.

This is an assay-indicated binding hit; no equilibrium affinity is reported. The displayed design is close to a natural sequence (87% identity over 86% coverage).

Fig. 6E–F, cropped; Fig. S19; calmodulin-design and spectroscopic Methods; Supplemental Table 1.

p53 scaffolds: binding survives relaxed structural filtering



1,300 generated: 900 Seq + 400 MSA.
24 nominated: 19 Seq + 5 MSA; all expressed in a cell-free system. MSA candidates failed

global pLDDT > 70, so selection used motif RMSD alone. BLI against full-length human

MDM2, triplicate:
8/24 show some binding;
4/24 have $K_D < 50$ nM in at least 2 replicates (1 Seq + 3 MSA). Two MSA hits: mean K_D

25.9 and 40.2 nM.

Fig. 6H–I, cropped; Figs. S20–S24; p53 selection/BLI Methods. Functional hits need not have high global pLDDT.

Strength of evidence: keep the denominator visible

Task	Selected tests	Direct conclusion
Unconditional	4/25 characterized	Some generated proteins express and have CD-supported secondary structure
Cox15 targeting	8/8 localize	Selected replacements preserve localization in one cellular context
Calcium scaffold	12/13 express; 1 binding-positive	Expression does not imply the intended binding function
p53–MDM2	8/24 any; 4/24 strong	Sequence-conditioned designs can yield measured binders

Each experiment tests a filtered pipeline. Different tasks use different selection rules and assays; the fractions are not comparable general success rates.

Figs. 2, 5, 6; selection Methods; Supplemental Table 1. Strong MDM2: $K_D < 50$ nM in ≥ 2 replicates.

Limitations that change the interpretation

1. **Objective versus flexibility.** OADM enables arbitrary conditioning, but LRAR is stronger on several unconditional metrics; ESM-MSA is competitive for scaffolding.
2. **Proxy metrics.** Predicted folds, embedding distances, disorder scores, and same-length Hamming distance are incomplete measures of function and novelty.
3. **Context and selection.** Homologs can anchor designs near known families; small filtered wet-lab sets do not establish general hit rates.

The 2026 public review also requests calibration against newer competitors such as DPLM and ProtBFN. Original baselines should not be read as a comprehensive 2026 ranking.

Tables S1, S3, S6; Methods; eLife public reviews #1–2. Reviewer concerns are distinct from new experimental evidence.

Discussion: what would make the next experiment decisive?

Working interpretation: EvoDiff's durable contribution is an experimentally supported framework for flexible sequence conditioning, including regions poorly represented by static structure.

1. Can a matched-budget, matched-selection comparison isolate the benefit of OADM over LRAR and MLM sampling?
2. Across unrelated IDR functions, how much comes from the generator, contextual homologs, and downstream property predictors?
3. Can iterative token revision or guidance increase functional hit rates without collapsing novelty?

Open models/code: github.com/microsoft/evodiff.
Full source and review: doi.org/10.7554/eLife.112029.1.

Discussion; eLife Reviewed Preprint v1. Questions are our proposed follow-ups, not reported results.